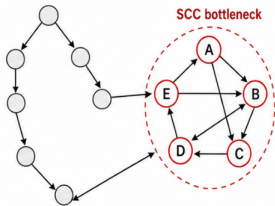
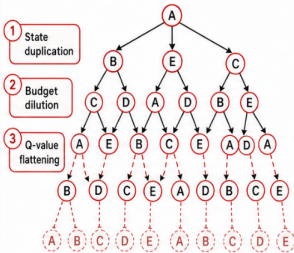


Physical Document Graph



- Same records can be revisited
- Evidence is cyclic
- Terminal depth is unclear

Vanilla TreeMCTS on the SCC



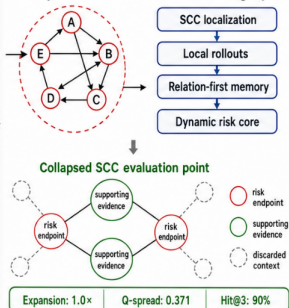
Expansion: 6.4×

Truncation: 100%

Q-spread: 0.044

SA-MCGS:

Collapse the SCC into a Risk Subgraph



C1

SCC-aware search

C2

Evidence memory

C3

Dynamic risk core